

Running In Circles

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One popular run on the Stanford campus is Campus Drive Loop. There are sidewalks on either side of the road, giving the runner a choice of which to take. On days when I felt low-energy, I would sometimes take the inside sidewalk, figuring that this would shorten my run. But by how much?

If the loop were a perfect circle, the answer would be easy – my savings would be 2π times the width of the road, implying a shorter route by something like 300 ft. But as shown in Figure 1, the loop is far from circular, and is not even convex. I wondered, is 300 feet still a reasonable answer, or is it possible that the savings could be much more or much less?

After drawing a few simple shapes (mostly compositions of straight lines and partial circles), I became convinced that in fact, the answer was 2π times the width of the road, regardless of its shape. But I felt I lacked the tools to tackle the problem formally. This is where my friend Jake Levinson comes in. I posed this question to him while on Safari in Tanzania, and he promptly tackled it with gusto (see Figure 2).

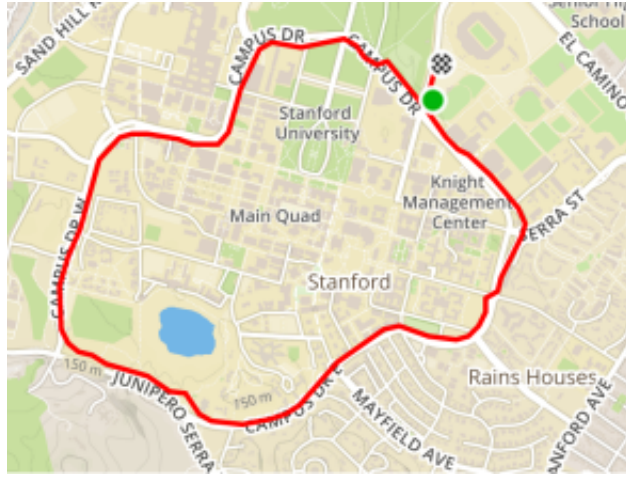


Figure 1: Campus Drive Loop at Stanford University.



Figure 2: Left: Jake sketching ideas in a cafe in Moshi. Right: how we spent the rest of our time.